

Part 1: Controller Area Network (CAN) Fundamentals

The Foundation of Automotive Embedded Communication

1. The Historical Automotive Problem

Prior to the 1980s, vehicle electrical components used point-to-point physical wiring. Every single switch was wired directly to its corresponding actuator. As safety and comfort features (ABS, cruise control, electronic ignition, central locking) proliferated, vehicle wire harnesses exceeded several kilometers in length and 100+ kg in weight, introducing severe reliability, maintenance, and space issues.

In 1986, Robert Bosch GmbH invented the **Controller Area Network (CAN)** to replace point-to-point wiring with a shared, high-reliability, two-wire digital bus.

2. Physical Layer: Differential Signaling

A CAN bus operates using two balanced, twisted copper wires: **CAN High (CAN_H)** and **CAN Low (CAN_L)**, terminated at both ends with 120-ohm resistors (giving an overall nominal bus impedance of 60 ohms).

- **Recessive State (Logical '1'):** Both CAN_H and CAN_L sit at approximately 2.5V (Differential voltage = 0V).
- **Dominant State (Logical '0'):** CAN_H is driven to ~3.5V and CAN_L is pulled to ~1.5V (Differential voltage = ~2.0V).

Because the signal is differential, any electromagnetic interference (EMI) induces equal noise voltages on both lines, which completely cancels out at the receiver's differential amplifier. This provides exceptional noise immunity in harsh automotive environments.

***Key Takeaway:** A dominant bit ('0') always overrides a recessive bit ('1') on the physical bus. This electrical property is the bedrock of CAN's non-destructive bus arbitration.*

3. Non-Destructive Bitwise Arbitration

Unlike Ethernet (CSMA/CD), where simultaneous transmissions result in collisions and lost packets requiring random backoff delays, CAN uses **lossless bitwise arbitration**:

1. Every node listens to the bus while transmitting bit-by-bit.
2. If Node A transmits a recessive '1' but senses a dominant '0' on the wire, it immediately knows a higher-priority node is speaking.
3. Node A backs off instantly and transitions to listening mode without disrupting the winning frame.
4. The highest-priority frame (the one with the lowest numerical CAN ID) continues uninterrupted.

4. Classical CAN Frame Structure

A standard CAN 2.0B frame comprises the following fields:

Field	Length	Description
SOF	1 bit	Start of Frame (dominant bit to synchronize clocks).
CAN ID	11 bits	Message identifier & priority indicator (e.g. 0x100).
RTR	1 bit	Remote Transmission Request (data vs request frame).
Control / DLC	6 bits	Data Length Code (specifies 0 to 8 bytes of data).
Data Field	0-8 bytes	Application payload (e.g., speed, torque, SoC).
CRC Field	16 bits	Cyclic Redundancy Check for error detection.
ACK Slot	2 bits	Transmitting node sends recessive; any receiver drives dominant to acknowledge.
EOF	7 bits	End of Frame (recessive bits).

5. Broadcast & Content-Addressed Model

In CAN, **there are no destination node addresses**. A transmitter broadcasts a message labeled by ID (e.g., ID 0x100 = 'Powertrain Speed & RPM'). All ECUs connected to the bus receive the message simultaneously. Each ECU configures hardware acceptance filters so it only interrupts the CPU for CAN IDs relevant to its function.